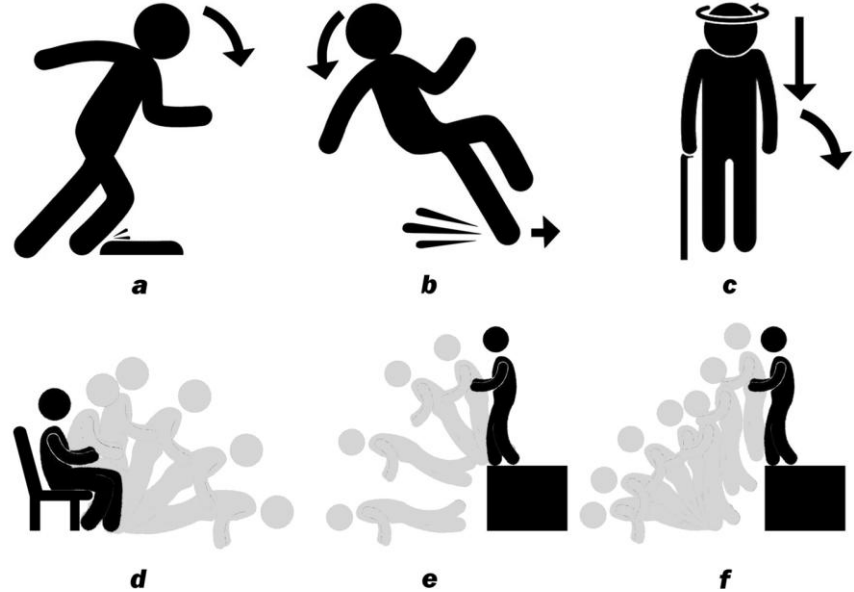


Smart Fall Detection Wrist Band

Elderly People

INTRODUCTION

A fall detection wristband is a wearable safety device that uses built-in motion sensors to automatically identify sudden drops and hard impacts. Upon detecting a fall, it triggers an emergency alert and shares the user's location with caregivers or medical services, helping to reduce response times and prevent severe injuries.



Problems Faced by Elderly People

👴 Falling while walking

😵 Dizziness

❤️ Low blood pressure

🦴 Bone fractures

🏠 Living alone

🚑 Delay in getting medical help



Objective of fall detection wrist band

The objective of a fall detection wristband:

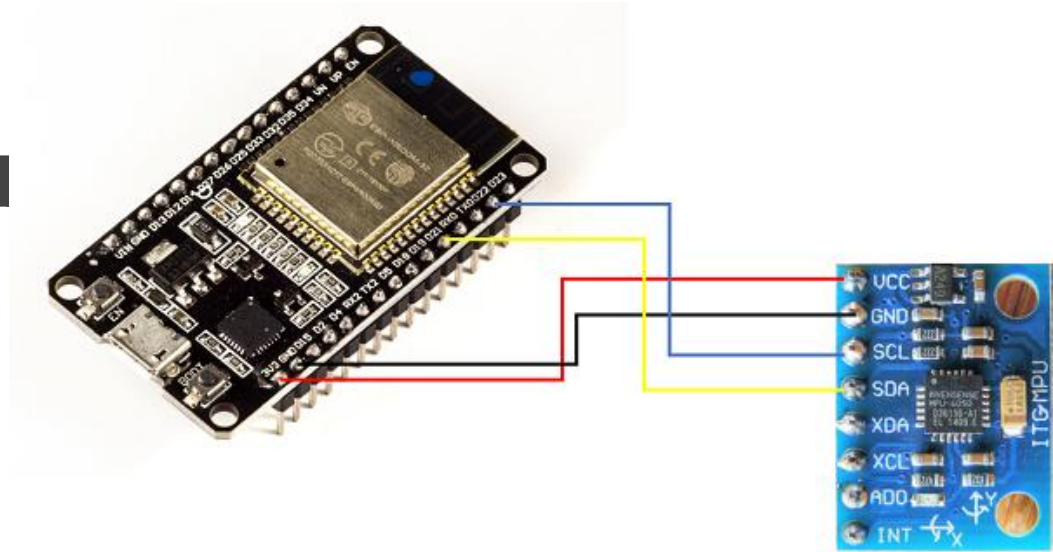
To automatically detect falls and provide immediate alerts or emergency assistance, helping reduce injury severity and improving safety, especially for elderly individuals, people with disabilities, or those with medical conditions.

Components Used

ESP8266(WIFI MODULE)

MPU6050 Sensor

Jumper Wires



Working Principle

Wear Wrist Band



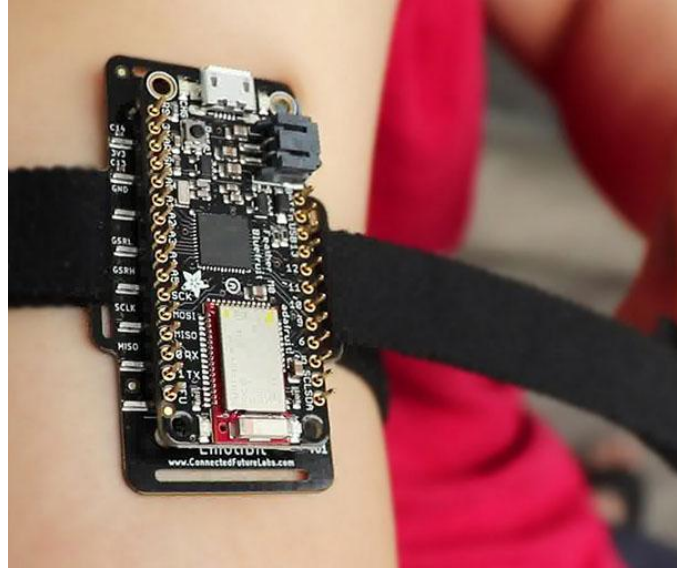
Sensor Detects Fall



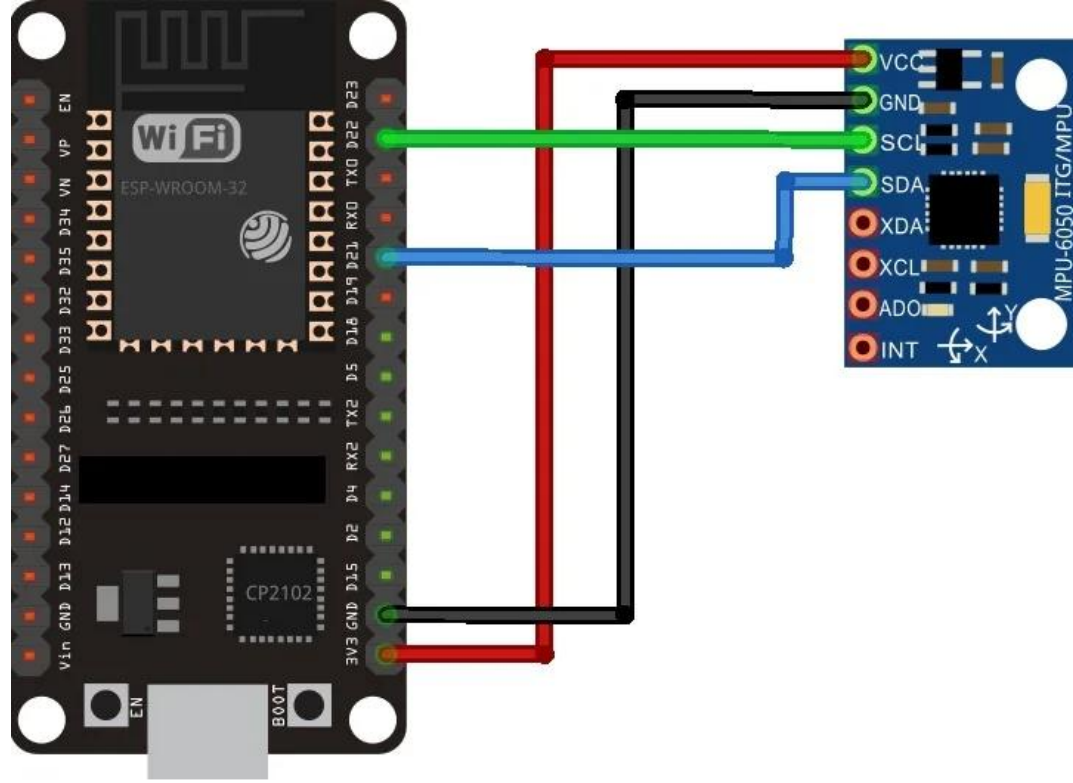
ESP8266 Processes Data



MPU 6050



Connection



Advantages

Advantages

Saves lives

Easy to wear

Low cost

Rechargeable

Quick emergency alerts



Advantages & Applications

Advantages

Saves lives

Easy to wear

Low cost

Rechargeable

Quick emergency alerts



Future Scope

Mention future improvements:

GPS location tracking

Heart-rate monitoring

SpO₂ monitoring

AI-based fall detection.

Voice assistant

Mobile app integration



FUTURE SCOPE

Smart **Fall Detection** Wrist Band for Elderly People

Innovating Today for a Safer Tomorrow

1. GPS LOCATION TRACKING



Real-time location tracking to find the elderly person anywhere instantly.



2. HEART RATE MONITORING

Continuous heart rate monitoring to detect irregularities and health issues.



3. SpO₂ MONITORING



Monitors oxygen saturation level to keep track of the user's health.

4. AI BASED FALL DETECTION

Use of AI & Machine Learning for more accurate fall detection and fewer false alerts.





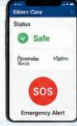
5. VOICE ASSISTANT



Voice assistant to help the elderly person call for help or perform tasks.

6. MOBILE APP INTEGRATION

Dedicated mobile app to monitor real-time data and receive instant alerts.



7. CLOUD DATA STORAGE



Stores health data in the cloud for long-term analysis and better healthcare decisions.

8. LONG BATTERY LIFE & ENERGY EFFICIENCY



Future versions will have longer battery life and better power management.

9. FALL ANALYTICS & REPORTS

Detailed analytics and reports for doctors and caregivers to improve elderly care.



The future is smarter, safer and more connected. Together, we can build a better tomorrow for our elders.

Conclusion

The smart fall detection wrist band for elderly people is a wearable device that improves the safety of senior citizens by detecting falls and sending emergency alerts to caregivers or family members it helps reduce response time, provides peace of mind, and supports independent living. this project is a simple, affordable, and effective solution that can help protect elderly people in everyday life.

Thank You